GB Flexible Generation Dashboard: Requirements

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Purpose

The dashboard shows when Great Britain still relies on flexible gas generation, so planners can see how often, how hard and in what conditions gas plants are called on as wind and solar grow.

The problem: gas fell from 44.8% of GB generation in 2009 to 26.7% in 2025, and coal has gone. But gas still covers the gaps: cold, dark, still evenings when wind and solar cannot meet demand. Average annual shares hide this. A planning team needs to see the hours, seasons and weather conditions when flexible generation is essential.

Scenario: this is a portfolio project built on real, published data from the National Energy System Operator (NESO). The stakeholder roles are typical of a flexible generation business such as SSE Thermal; no real individuals are represented.

Stakeholders and users

Three groups use the dashboard, each with a different question about flexible generation.

User	What they need
Commercial and strategy lead (sponsor)	Long-term shift in the generation mix and the role gas still plays
Operations and maintenance planner	When gas plants run hardest, so maintenance can be planned for low-demand periods
Sustainability and carbon reporting	How carbon intensity changes with the mix, and when the grid is cleanest

Business questions

The dashboard must answer six questions; every page traces back to at least one.

1. **How has the generation mix changed since 2009,** and how much of GB electricity does gas still supply?
2. **When is gas needed most:** by season, by month and by time of day?

3. **How does gas respond when the wind drops?** How much more gas runs in the least windy periods?
4. **How hard does gas have to work at its peak,** and how often does it supply more than half of GB electricity?
5. **How does carbon intensity change** with the generation mix, and when is the grid cleanest?
6. **Can the data be trusted?** Which readings are incomplete or implausible?

Data source and scope

One dataset from NESO, the Historic GB Generation Mix, covers every half-hour from 1 January 2009 to 4 November 2026: 312,793 readings.

Field group	What it holds	Unit
Generation by fuel	Gas, coal, nuclear, wind (transmission), embedded wind, hydro, imports, biomass, solar, storage, other	MW
Totals	Total generation; low-carbon, zero-carbon, renewable and fossil totals	MW
Shares	Each fuel and total as a share of generation	%
Carbon intensity	Carbon dioxide emitted per unit of electricity	gCO_2/kWh

Source: NESO Historic Generation Mix.

In scope: Great Britain as a whole, half-hourly, 2009 to date; dashboard pages default to the last full year (2025) where a single year is shown.

Out of scope: regional or plant-level data, prices and costs, demand forecasts, and Northern Ireland.

Two rules:

- **Wind means wind plus embedded wind.** Embedded wind is connected to local networks and is part of the wind picture.
- **Shares are calculated from summed MW,** never by averaging half-hourly percentages.

KPI definitions

Each KPI is calculated from summed half-hourly MW over the period shown, so busy half-hours count for more than quiet ones.

KPI	Definition	Calculation	Answers question
Gas share	Share of GB generation supplied by gas	Sum of GAS ÷ sum of GENERATION	1, 2
Fuel share	Share supplied by each fuel	Sum of fuel MW ÷ sum of GENERATION	1
Wind share	Wind including embedded wind	(Sum of WIND + WIND_EMB) ÷ sum of GENERATION	1, 3
Peak gas output	Highest half-hourly gas output in the period	Maximum of GAS (MW)	4
Gas-dominant half-hours	How often gas supplies most of GB electricity	Share of half-hours where GAS_perc is above 50%	4
Gas share by wind level	Gas share in each tenth of half-hours, ranked from least to most windy	Gas share within each wind decile	3
Average carbon intensity	Typical carbon emitted per unit of electricity	Mean of CARBON_INTENSITY (gCO_2/kWh)	5
Flagged readings	Half-hours failing a data quality check	Count of rows flagged	6

Rules: shares are never averaged; flagged readings are excluded from every KPI; time-of-day analysis uses UK local time, converted from the UTC timestamps in the source.

Dashboard pages and requirements

Four pages, each answering a set of the business questions, with the overview first.

Page	What it shows	Answers question
Overview	Gas, wind and carbon headline figures for the selected year, and the mix since 2009	1, 5
When gas is needed	Gas share by month and by hour of day, with peak output and gas-dominant half-hours	2, 4

Page	What it shows	Answers question
Gas and wind	Gas share across wind deciles, and how gas rises as wind falls	3
Data quality	Readings checked, anomalies found and how each is handled	6

Functional requirements

- [ ] A year filter on every page, defaulting to the last full year
- [ ] Hover on any chart to read exact values
- [ ] Plain-English titles and short notes explaining how to read each chart
- [ ] Consistent fuel colours across all charts
- [ ] The source and data date shown on every page
- [ ] Works on desktop and mobile

Data quality, refresh and acceptance

The first load found three issues; each is flagged and handled transparently, never silently changed.

Check	Result on first load	Handling
Missing or duplicate half-hours	None in 312,793 readings	No action
Fuels add up to total generation	48 half-hours differ by up to 3 MW	Rounding; accepted
Impossible values	7 July 2026, 10:00 and 10:30: gas at 0 MW and zero-carbon share at 104.1%	Flagged and excluded
Coal after GB coal closure (September 2024)	24 half-hours show 5 to 198 MW of coal	Flagged for investigation; shown separately

Refresh: NESO updates the dataset continuously; the dashboard reloads the full file and reruns every check on each refresh.

Acceptance criteria

- [ ] Annual shares match those calculated directly from the source file for three sample years

- [ ] Every flagged reading is listed on the Data quality page with its reason
- [ ] Changing the year filter updates every chart on the page
- [ ] Each of the six business questions can be answered from the dashboard